

VALLEY VIEW UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE



SENIOR RESEARCH PROJECTS

[SUBMISSION GUIDELINES]

June 2016

A. INTRODUCTION

The department has adopted two categories of research projects which we call

- i. Research and System Development
- ii. Research Only

REQUIRED CHAPTERS

▪ Research and System Development

Required Chapters

1. General Introduction (Research Proposal)
 2. Literature Review
 3. Crystallization of the Research Problem
 4. Analysis of the Proposed System
 5. Detailed design of the Proposed System
 6. System Implementation and Testing
 7. System Documentation
 8. Conclusion and Recommendations
- References

▪ Research Only

Required Chapters

1. General Introduction (Research Proposal)
 2. Literature Review
 3. Research Methodology
 4. Findings and Discussions
 5. Conclusion and Recommendations
- References

B. DOCUMENT DETAILS (Research and System Development)

Cover Page

Table of Contents

Acknowledgements

Declaration

Abstract

1. General Introduction (Research Proposal)

1.0 Introduction and Opportunity Statement

1.1 Subject and Field of Study

1.2 Study Objectives

1.2.1 Global (General) Objectives

1.2.2 Specific Objectives

1.3 Background to the Study

1.4 Scope of Study

1.5 Justification of the Study/Significance of Study

1.6 Methodology (*Interview, Questionnaire, Observation, Desktop Review*)

Please Note: For Research + System Development, specify and justify the Software Development Process Model to be employed.

1.7 Expected Results of the Study & Possible Use (Research/Study Deliverables)

1.8 Presentation of Thesis (Chapter-by-Chapter Summary of Thesis)

1.9 Study Work Plan (Timelines) **[10 Marks]**

2. Literature Review

You are required to conduct a critical and in depth evaluation of what have been published on your chosen research area/topic. The purpose of this exercise is to summarize, synthesize, analyze and evaluate the arguments of others to support your own argument. Therefore, describe and analyze the knowledge that exists and identify the gaps in research related to your project topic. This should clarify the relationship between your research topic and the work that has previously been done.

Also, you are required to review at least two existing computer-based systems or similar systems for those undertaking Research and System Development.

Please note: Referencing websites should not exceed **30%** of the total references.

3. Crystallization of the Problem

You are required to undertake a study of the system implemented by your case study organization(s). Provide information on the following

- i. Background/History
- ii. How they operate (components/features of the system they use even if it is manual, processes or procedures involved). Include the system context diagram and the process or data flow diagrams.
- iii. The strength/advantages and the weaknesses/disadvantages of the system they use.

- iv. Conduct a comparative study by juxtaposing with what exist in the body of knowledge

Please note: The list above is not exhaustive. You can discuss with your supervisor on additions to include.

4. Analysis of the Proposed System

A typical analysis of the proposed system has the following subheadings;

4.0 Overview of the proposed system

- Functional Requirements
- Non-functional Requirements

4.1 Major features/components of the proposed system

4.2 Benefits/Advantages of the proposed system

4.3 System Context Diagram of the proposed system

Please note: The list above is not exhaustive. You can discuss with your supervisor on additions to include.

5. Detailed design of the Proposed System

5.0 Functional Processes of the Proposed System

5.1 Algorithms or flowchart of the System

5.2 Process Models

Data Flow Diagrams Levels 0, 1, 2 and 3 where applicable

Decision Trees

UML Class Diagrams and Sequence etc.

Use Cases or User Scenarios or Activity Diagram

5.3 Data Models

Data Dictionary

-Database Schema

-Tables

-Table Relationship Diagram

-You may use Entity relationship diagrams

5.4 Model the Views as well with designs to provide snapshots

5.5 Show interactivity with Collaboration, Class Sequencing, State diagram, IPO and HIPO charts to show coupling between modules

6. Implementation of System design and Testing

6.0 Implementation/the Creation of the Software/Methods used in the project construction - This should give a vivid description of how the detailed design was implemented

6.1 This chapter shows or gives a summary how the entire system or project was carried out. Students should take time to indicate step by step process used in the construction of the software or research.

6.2 Detailed semantics of the operations of the software must be outlined with snippets of codes to illustrate techniques and strategies adopted/adapted for implementation

6.3 The purpose of the implementation is to give the technical details so one can recreate the system by following the description

6.4 Testing - Description of the testing strategy - Statement of Test Cases for the proposed system - Indicate how the test cases were tested and their results
Conjecturing test cases will not be tolerated at all

7. System Documentation

- User Manual for the system or a multimedia guided tour will be appropriate
- Description of both the Hardware and Software required for the implementation (installation) of the proposed system is appropriate here
- Provide both text and snapshots of how to use the various components of the system. The purpose is to enhance usability and learnability of the system developed

8. Conclusion, Recommendation and Further Work

C. DOCUMENT DETAILS (Research Only)

Cover Page

Acknowledgements

Declaration

Abstract

Table of Contents

1. General Introduction (Research Proposal)

- 1.0 Introduction
- 1.1 Subject and Field of Study
- 1.2 Study Objectives
 - 1.2.1 Global (General) Objectives
 - 1.2.2 Specific Objectives
- 1.3 Background to the Study
- 1.4 Scope of Study
- 1.5 Justification of the Study/Significance of Study
- 1.6 Methodology (*Interview, Questionnaire, Observation, Desktop Review*)
- 1.7 Expected Results of the Study & Possible Use (Research/Study Deliverables)
- 1.8 Presentation of Thesis (Chapter-by-Chapter Summary of Thesis)
- 1.9 Study Work Plan (Timelines)

2. Literature Review

You are required to conduct a critical and in depth evaluation of what have been published on your chosen research area/topic. The purpose of this exercise is to summarize, synthesize and analyze the arguments of others to support your own argument. Therefore, describe and analyze the knowledge that exists and identify the gaps in research related to your project topic. This should clarify the relationship between your research topic and the work that has previously been done.

Please note: Your website references should be less than **30%** of the total references.

3. Research Methodology

- 3.1 Type of Research
- 3.2 Population
- 3.3 Sample
- 3.4 Sampling technique
- 3.5 Data collection method
- 3.6 Type of data to be collected
- 3.7 Instrument for data collection
- 3.8 Instrument validity and reliability
- 3.9 Instrument structure to meet research objectives
- 3.10 Method of data analysis
- 3.11 Instrument (*i.e. Questionnaire, experiment design, interview questions, etc*)
- 3.12 Computer Simulation of Instrument

4. Data Presentation, Findings and Discussions
5. Design and Implementation
6. Conclusion and Recommendations

D. REFERENCES/BIBLIOGRAPHY

You are required to follow the format below

- **Books**
Author(s), *Title*, Publisher(s), Edition, Year of Publication
Example: Quarshie H, *Computer Science Research Made Simple*, 1st ed, 2011
- **Journals Article**
Author(s), *Title*, Journal Name, Year of Publication, Pages
Example: Quarshie H, *Computer Science Research Made Simple*, Journal for Advances in Computer Technology, 2011, Pages (34 – 37)
- **Conference Proceedings/Report**
Author(s), *Title*, Pages, Year of Publication, Name of Conference, City, Publisher(s)
Example: Quarshie H, Damoah D., *Computer Science Research Made Simple*, Pages (34 – 37), 2011, Ghana Researchers Conference, Koforidua, Silverplatter Ventures
- **Document/information from Website**
Author(s), Name of Web Page, Name of Website, Year of Publication, *URL*
Example: Silverplatter Ventures, What is Literature Review, Experiment Resources, 2011, <http://www.experiment-resources.com/what-is-a-literature-review.html>
- **Interview**
Interviewee, Title, Interviewer, Date
Example: Mansa Mansotwenee, Operations of Silverplatter Ventures, Quarshie H, 11-11-2011

NOTE: Sort your references in alphabetical order.

E. GENERAL DOCUMENT FORMAT

The general format of the document you present are as follows;

- **Font Style:** - Courier New
 - **Font Size:** - 12
 - **Paragraph Line Spacing:** - Double Line spacing
 - **Margins:** - 1 inch for Top, Bottom, Left, and Right
 - **Text Alignment:** - Justify
 - **In text referencing:** - APA Format
 - **Page Numbers:** - Should be centered at the bottom page. Avoid the use of page numbers such as
-
- 7 | Page
- No footers or headers.
 - Labels for all figures should be at the top of the figure and the figure should be broad enough (i.e. cover at least ½ of the page).
 - **Please Note:** We will look out for the general presentation layout, and discussion flow of the report.
 - You should have ONLY one introduction, and that should be in Chapter 1.

F. GRADING SCHEDULE

**VALLEY VIEW UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE
FINAL YEAR PROJECT 1 GRADING SCHEDULE**

STUDENT NAME:

STUDENT ID #:

PROJECT TITLE:

DESCRIPTION	MARKS ALLOTTED	MARKS ASSIGNED
<u>Defining the purpose</u> of the research, justifying the research work and clearly stating how the research was conducted and the summary of the presentation of the research report	10 Marks	
<u>Literature Review</u> - The extent, scope, details and relevance of the Literature Review of previous research and other works conducted in the area of the project.	30 Marks	
<u>Crystallization of the Problem</u> – Demonstration of the understanding of the problem to be solved.	20 Marks	
<u>Analysis</u> - The relevance, scope, and the technical details, the suitability of the adopted methodology and demonstration of the student's comprehension of the various technical, conceptual and methodological issues pertaining to the analysis conducted.	20 Marks	
<u>General Presentation</u> laid out, discussion flow of the report.	10 Marks	
<u>Oral Presentation</u>	10 Marks	
TOTAL	100	

Name of Supervisor/Moderator: Signature:

Grade: Date:

**VALLEY VIEW UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE
FINAL YEAR PROJECT 11 GRADING SCHEDULE**

STUDENT NAME:

STUDENT ID#:

PROJECT TITLE:

DESCRIPTION	MARKS ALLOTED	MARKS ASSIGNED
<u>Design</u> – The relevance, scope, and the technical details, the suitability of the adopted design tools used and demonstration of the student’s comprehension of the various technical, issues pertaining to the design of the proposed system. The relevance and completeness of the system design outputs.	50	
<u>Implementation</u> – The extent to which the implemented system addresses the key specific objectives of the project. The demonstration of the working features of the developed system in terms of sample data/information. The quality and the relevancy of the user manual produced. The system documentation and the evidence of testing of the system.	25	
<u>General Presentation</u> laid out discussion flow of the report.	15	
<u>Oral Presentation</u>	10	
<u>Total</u>	100	

Name of Supervisor/Moderator: Signature:

Grade:

Date:

APPENDIX A – Cover Page Sample

VALLEY VIEW UNIVERSITY [line 1]
FACULTY OF SCIENCE [line 2]
DEPARTMENT OF _____ [line 3]

A RESEARCH PROJECT SUBMITTED IN PARTIAL FULFILLMENT OF [line 11]
THE REQUIREMENTS FOR THE BACHELOR OF SCIENCE (BSC.) IN
_____ DEGREE

TOPIC: [line 20]
[sample] **A COMPARATIVE STUDY OF THE EFFECT OF Facebook ON**
THE FACIAL EXPRESSIONS OF COMPUTER SCIENTISTS

BY: [line 30]
KWAME MUSA MANSOTWENEE

ID: [line 38]
299CT01067071

SUPERVISOR: [line 43]
PROF. PEACE ASOMDWEHENE

DATE: [line 50]
MAY 2012

Please Note:

- **Text Alignment:** Centered
- **Paragraph Line Spacing:** - Double Line spacing
- **Font Size:** 12
- **Font Style:** Courier New
- **Bold** the entire text for the cover page

- No underline or itaclize

APPENDIX B: Sample Declaration

DECLARATION

This is to declare that, the research work underlying this senior research project has been carried out by the under mentioned student under the supervisor. Both student and the supervisor certify that the work documented in this thesis is the output of the research conducted by the student as part of her final year project work in partial fulfillment of the requirement of the Bachelor of Science in _____ degree.

STUDENT:

Kwaku Troublesome

SIGNATURE

DATE:

SUPERVISOR:

Prof. Chaos Peace

SIGNATURE

DATE:

Note:

Candidates must submit a master copy of the project work, including signed approval page, to the secretary at least two weeks before defence. From this master copy, four copies will be made for the University. At this point, the department will notify the Academic Records Office that you have completed your project work. This must take place no later than noon on Friday, at least two weeks preceding graduation. (Your master copy will be returned to you.)